$$W_{1,2}(X,P,t) = N^2\delta^2\frac{\delta_2}{\delta_1}\exp\left[-\frac{(X\mp X_c)^2}{\delta_1^2}\right]$$
$$\times\ \exp\left[-\delta_2^2\left(P\mp P_c-\beta(X\mp X_c)\right)^2\right], \tag{5.72a}$$
$$W_{\rm int}(X,P,t) = 2N^2\delta^2\frac{\delta_2}{\delta_1}\exp\left(-A_{\rm int}\right)\exp\left[-\frac{X^2}{\delta_1^2}-\delta_2^2(P-\beta X)^2\right]$$
$$\times\ \cos\left[2\kappa_p P+2(\kappa_x-\beta\kappa_p)X\right]. \tag{5.72b}$$